

LECTURE 16: DIFFERENCE IN DIFFERENCES

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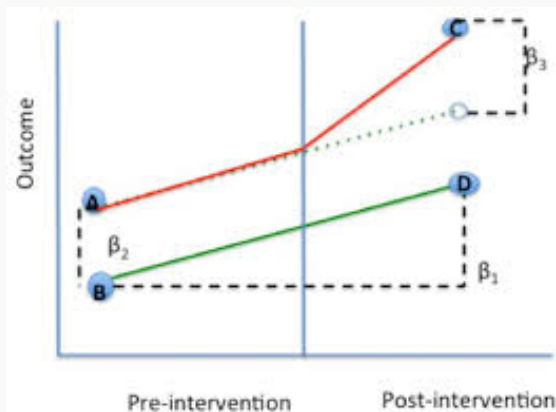
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INTRODUCTION

- In this lecture we discuss **Difference-In-Differences** (DID) methods. The set up is that there are two or more groups in a population, with units observed in two or more periods. In some periods and some groups are exposed to a new regime that we wish to evaluate.
- The two concerns are
 - the comparison of different groups at the same point in time is potentially biased because the groups may differ systematically in unobserved ways.
 - The comparison of the same group at different points in time is potentially biased because other changes may occur over time.
- Under some assumptions we can compare the change in the group that was exposed to the new regime in the second period to the change in the group that was never exposed to the new regime and eliminate both biases.

DIFFERENCE IN DIFFERENCES PLOT



TWO GROUPS AND TWO PERIODS WITH REPEATED CROSS-SECTIONS

- There are three different data configurations we consider, with somewhat vaguely defined boundaries.
 - **Proper panel data**: we observe the outcome Y_{it} for N units and T time periods, $i = 1, \dots, N$ and $t = 1, \dots, T$. For example, Lalonde data, where we observe outcomes for individuals in three periods.
 - **Repeated cross-sections**, we observe Y_i , for unit $i = 1, \dots, N$, in period T_i (different for different individuals), with individual i belonging to group G_i , and G_i is binary, treatment and control group. For example, Meyer et al, where we observe individual who got injured on the job, in one of two periods, in one of two groups (low and high earners).
 - **Repeated cross-sections with multiple groups**. Now $G_i \in \{1, \dots, G\}$, split into treatment and control groups. For example Bertrand, Duflo and Mullainathan, who observe individuals in 50 states, divided into control and treated states.

REPEATED CROSS-SECTIONS

- For the repeated cross-section setting the standard model for the DID design is as follows.
 - Individual i belongs to a group, $G_i \in \{0, 1\}$ (where group 1 is the treatment group),
 - is observed in time period $T_i \in \{0, 1\}$. Letting the outcome be Y_i ,
 - the observed data are the triple (Y_i, G_i, T_i) .
- Using the potential outcome notation, let $Y_i(C)$ denote the outcome for individual i if that individual does not receive the treatment, and let $Y_i(T)$ be the outcome for the same individual if he or she does receive the treatment.
- $I_i = G_i T_i$ is an indicator for the treatment
- The realized (observed) outcome for individual i is

$$Y_i \equiv Y_i(C)(1 - I_i) + I_i Y_i(T).$$

REPEATED CROSS-SECTIONS

- In the standard DID model the outcome for individual i in the absence of the intervention satisfies

$$Y_i(C) = \alpha + \beta T_i + \gamma G_i + \varepsilon_i.$$

- The second parameter, β , represents the time component. The third parameter, γ , represents a group-specific, time-invariant component. The third term, $\varepsilon_i \perp (G_i, T_i)$, represents unobservable characteristics of the individual.
- The standard DID estimand is

$$\begin{aligned} \tau^{DID} = & \left[\mathbb{E}[Y_i | G_i = 1, T_i = 1] - \mathbb{E}[Y_i | G_i = 1, T_i = 0] \right] \\ & - \left[\mathbb{E}[Y_i | G_i = 0, T_i = 1] - \mathbb{E}[Y_i | G_i = 0, T_i = 0] \right]. \end{aligned}$$

REPEATED CROSS-SECTIONS

- The interpretation of the standard DID estimand depends on assumptions about how outcomes are generated in the presence of the intervention.
- It is often assumed that the treatment effect is constant across individuals, so that $Y_i(T) - Y_i(C) = \tau$.
- Combining this restriction with the standard DID model for the outcome without intervention this leads to a model for the realized outcome

$$Y_i = \alpha + \beta T_i + \gamma G_i + \tau I_i + \varepsilon_i.$$

- More generally, the effect of the intervention might differ across individuals. Then, the standard DID estimand gives the average effect of the intervention on the treatment group.

REPEATED CROSS-SECTIONS

- Given the above model for the observed outcomes we can use Ordinary Least Squares to estimate the estimand of interest:

$$\min_{\alpha, \beta, \gamma, \tau} \sum_{i=1}^N \left(Y_i - \alpha - \beta T_i - \gamma G_i - \tau I_i \right)^2$$

Simple algebra shows that this leads to

$$\hat{\tau} = \bar{Y}_{11} - \bar{Y}_{10} - (\bar{Y}_{01} - \bar{Y}_{00}),$$

where

$$\bar{Y}_{gt} = \frac{1}{N_{gt}} \sum_{i: G_i=g, T_i=t} Y_i,$$

is the average outcome in time period t and group g , and

$$N_{gt} = \sum_{i=1}^N \mathbf{1}_{G_i=g, T_i=t},$$

is the number of observations in time period t and group g .

REPEATED CROSS-SECTIONS

- If we assume that the residuals are homoskedastic, then we can use standard OLS results to get the variance of the DID estimator,

$$V(\tau)_{\text{homosk}} = \sigma^2 \left(\frac{1}{N_{11}} + \frac{1}{N_{10}} + \frac{1}{N_{01}} + \frac{1}{N_{00}} \right).$$

Using the DID representation the robust (Eicker-Huber-White) variance would be

$$V(\tau)_{\text{heterosk}} = \frac{\sigma_{11}^2}{N_{11}} + \frac{\sigma_{10}^2}{N_{10}} + \frac{\sigma_{01}^2}{N_{01}} + \frac{\sigma_{00}^2}{N_{00}},$$

where the within group/time-period variance σ_{gt}^2 can be estimated as

$$\widehat{\sigma_{gt}^2} = S_{gt}^2 = \frac{1}{N_{gt} - 1} \sum_{i: G_i=g, T_i=t} (Y_i - \bar{Y}_{gt})^2.$$

REPEATED CROSS-SECTIONS

- We can generalize this model if we have exogenous/pre-treatment covariates in a natural way.
- Suppose for unit i we observe X_i , and we now assume the structure

$$Y_i = \alpha + \beta T_i + \gamma G_i + \tau I_i + \delta' X_i + \varepsilon_i,$$

with the assumption that ε_i is independent of group/time dummies and of the covariates.

- Then we can again use OLS to estimate all parameters, and estimate the variance using least squares results either assuming homoskedasticity or allowing for heteroskedasticity.
- More generally we can exploit the presence of covariates in more flexible ways, interacting them with both the group and time dummies as well as with the treatment indicator.

MEYER-VISCUSI-DURBIN DATA

- Let us look at the basic DID set up with repeated cross-section data with the Meyer-Viscusi-Durbin (MVD) data.
- MVD look at the effect of a change in the benefit system that affects compensation in case of work related injuries on durations of disability spells.
- The benefit increase applied to high-earning but **not** to low-earning workers.
- The **outcome variable** is the number of weeks a worker spent on disability.
- MVD have **repeated cross sections** of individuals who were injured prior to, and after, the change in the benefit system, not a proper **panel**.
- The group sizes are $N_{00} = 1705$, $N_{01} = 1527$, $N_{10} = 1233$, and $N_{11} = 1161$.

MEYER-VISCUSI-DURBIN DATA: AVERAGES BY GROUP AND TIME PERIOD

		Period		Dif
		before	after	
Group	0	6.27	7.04	0.77
	1	11.18	12.89	1.72
	Dif	4.91	5.86	0.95

- The DID estimate is **0.95** with a robust standard error of 1.28.
- The OLS regression leads to

$$\widehat{\text{duration}}_i = \underset{(0.52)}{6.27} + \underset{(0.81)}{4.91 \times \text{gr}_i} + \underset{(0.76)}{0.77 \times \text{ti}_i} + \underset{(1.6)}{0.95 \times \text{gr} \times \text{ti}_i}$$

PROPER PANEL DATA

- Now suppose we have **panel** data where we observe the **same units at multiple points in time** (initially just two periods).
- Here we have N individuals, indexed $i = 1, \dots, N$, for whom we observe $(G_i, Y_{i0}, Y_{i1}, W_{i0}, W_{i1})$,
- $G_i \in \{0, 1\}$ denotes the group.
- In the standard DID case $W_{i0} = 0$, and $W_{i1} = G_i$, so only the treatment group is treated, and only in the last period.

PROPER PANEL DATA

- For estimation we can do the exact same thing as before. We can postulate a model

$$Y_{it} = \alpha + \beta t + \gamma G_i + \tau t G_i + \tau W_{it} + \varepsilon_{it},$$

and we can estimate this by OLS under the assumption that ε_{it} is independent of W_{it} , the group indicator and of each other. In the case without covariates this leads to the exact same double difference as before.

PROPER PANEL DATA

- We can also estimate this through ols for the regression function

$$Y_{it} = \mu + \alpha_i + \beta_t + W_{it}\tau + \varepsilon_{it}$$

as a two-way-fixed-effect regression (normalizing one of the α_i and one of the β_t to zero), leading to

$$\min_{\mu, \alpha, \beta, \tau} \sum_{i=1}^N \sum_{t=1}^T \left(Y_{it} - \mu - \alpha_i - \beta_t - \tau W_{it} \right)^2$$

- We can also run the regression

$$\dot{Y}_{it} = \dot{W}_{it}\tau + \eta_{it}$$

where we doubly-demean the variables

$$\dot{Y}_{it} = Y_{it} - \frac{1}{T} \sum_{s=1}^T Y_{is} - \frac{1}{N} \sum_{j=1}^N Y_{jt} + \frac{1}{NT} \sum_{s=1}^T \sum_{j=1}^N Y_{js}$$

and similar for $\dot{W}_{it}$.

UNCONFOUNDEDNESS *versus* DIFFERENCE-IN-DIFFERENCES

- Note that a very different approach with panel data would be to assume **unconfoundedness** given lagged outcomes.
 - The **two-way-fixed-effect** approach suggests the regression

$$Y_{i1} - Y_{i0} = \alpha + \tau G_i + \varepsilon_i.$$

- The **unconfoundedness** assumption would suggest

$$Y_{i1} - Y_{i0} = \alpha + \beta Y_{i0} + \tau G_i + \eta_i.$$

- It is **not** the case that the unconfoundedness assumption is less restrictive by freeing up the coefficient on the lagged value of the outcome. The DID assumption implies that adjusting for lagged outcomes in fact compromises the comparison because Y_{0i} may in fact be correlated with ε_i . In the end the two approaches make **different** assumptions.

UNFOUNDEDNESS *versus* DIFFERENCE-IN-DIFFERENCES

- Unfoundedness *versus* Fixed Effects
- Suppose in PA:

$$\begin{pmatrix} Y_{i0} \\ Y_{i1} \end{pmatrix} | PA \sim \mathcal{N} \left(\begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix} \right)$$

- in NJ:

$$\begin{pmatrix} Y_{i0} \\ Y_{i1} \end{pmatrix} | NJ \sim \mathcal{N} \left(\begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix} \right)$$

$$E[Y_{i1} | Y_{i0}, PA] = 1 + \rho Y_{i0}, \quad E[Y_{i1} | Y_{i0}, NJ] = 2 + \rho(Y_{i0} - 1)$$

$$Y_{i1} - Y_{i0} | PA \sim \mathcal{N}(1, 1 - \rho^2) \quad Y_{i1} - Y_{i0} | NJ \sim \mathcal{N}(1, 1 - \rho^2)$$

UNFOUNDEDNESS *versus* DIFFERENCE-IN-DIFFERENCES

- Unfoundedness:

$$E[Y_{i1} - Y_{i0} | Y_{i0}, \text{State}] = 1 + (\rho - 1)Y_{i0} + (1 - 2\rho) \times NJ_i$$

leading to

$$\tau_{\text{unconf}} = 1 - 2\rho$$

- TWFE:

$$E[Y_{i1} - Y_{i0} | \text{State}] = 1$$

leading to

$$\tau_{\text{twfe}} = 0$$

CARD-KRUEGER MINIMUM WAGE DATA

- These data are from the Card-Krueger paper on the effect of a change in the minimum wage in New Jersey.
- Card and Krueger collected data on employment at fast food restaurants in New Jersey and Pennsylvania before and after the change in the minimum wage.
- The data are converted to fulltime employment, counting parttime employees as half.
- The group sizes are $N_{PA} = 76$ and $N_{NJ} = 315$.

CARD-KRUEGER DATA: MEANS BY GROUP AND TIME PERIOD

		Period		Dif
		before	after	
Group	PA	20.01	17.52	-2.49
	NJ	17.05	17.50	0.45
	Dif	-2.97	-0.02	2.94

CARD-KRUEGER DATA: ESTIMATES

- The DID estimate is 2.94 with a robust standard error of 1.32.
- The OLS regression leads to

$$\widehat{\Delta \text{empl}}_i = -2.49 + 2.94 \times \text{group}_i$$

(1.01) (1.12)

again with a slightly smaller standard error than the we get under the robust variance calculation.

- An alternative regression here is that of the second period employment on the group dummy and the first period employment. This leads to

$$\widehat{\Delta \text{empl}}_i = 7.77 + 1.42 \times \text{group}_i - 0.51 \times \text{empl}_{\text{before},i}$$

(1.15) (0.94) (0.04)

MULTIPLE GROUPS AND MULTIPLE TIME PERIODS

- basic twfe model for panel data, multiple periods

$$Y_{it} = \alpha_i + \beta_t + \tau W_{it} + \varepsilon_{it}$$

- Bertrand, Duflo and Mullainathan (2004) were concerned about the case with **multiple groups** as well as **multiple time periods**. In that case there may well be correlation over time in the errors ε_{it} , leading to a multi-period version of the **Moulton** clustering problem.
- BDM show that ignoring this autocorrelation can have big effects on the properties of standard procedures such as OLS, and may lead researchers to erroneously conclude that there are effects of policy changes. See also Hansen (2006).
- Note that this is specific to the proper panel case (or the repeated cross-section case with grouped data)

MULTIPLE GROUPS AND MULTIPLE TIME PERIODS

- Repeated cross-sections, with group structure. Think observations on individuals, living in different states, observed in different periods. Each individual is observed only in one period.
- Now let there be $S + 1$ time periods, $T_i = 0, 1, \dots, S$, and $K + 1$ groups, $G_i = 0, 1, \dots, K$.
- The model we use is

$$Y_i = \beta_{T_i} + \gamma_{G_i} + \tau_{l_i} + \varepsilon_i = \sum_{s=0}^S \beta_s \mathbf{1}_{T_i=s} + \sum_{g=0}^K \gamma_g \mathbf{1}_{G_i=g} + \tau_{l_i} + \varepsilon_i.$$

MULTIPLE GROUPS AND MULTIPLE TIME PERIODS

- The error structure BDM and Hansen suggest is

$$\varepsilon_i = \eta_{G_i, T_i} + \nu_i,$$

- This allows for a idiosyncratic individual level error term ν_i , and a group-specific time-varying component η_{gt} .
- The unit level error term has the properties

$$\mathbb{E} [\nu_i \nu_j] = \begin{cases} \sigma_\nu^2 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

MULTIPLE GROUPS AND MULTIPLE TIME PERIODS

- The second group/time error component η_{gt} is allowed to be correlated over time with an autoregressive structure:

$$\eta_{gt} = \alpha\eta_{gt-1} + \omega_{gt},$$

- Here

$$\omega_{gt} \perp \eta_{gt-1}, \quad \omega_{gt} \perp \omega_{gs}, s \neq t, \quad \mathbb{E} \left[\omega_{gt}^2 \right] = \sigma_{\omega}^2.$$

- Then, with $\sigma_{\eta}^2 = \mathbb{E}[\eta_{gt}^2] = \sigma_{\omega}^2/(1 - \alpha^2)$, we have

$$\mathbb{E} [\eta_{gt}\eta_{hs}] = \begin{cases} \sigma_{\eta}^2 & \text{if } g = h, t = s, \\ \sigma_{\eta}^2 \alpha^{|t-s|} & \text{if } g = h, t \neq s, \\ 0 & \text{if } g \neq h. \end{cases}$$

MULTIPLE GROUPS AND MULTIPLE TIME PERIODS

- This implies that

$$\mathbb{E}[\varepsilon_i \varepsilon_j] = \begin{cases} \sigma_v^2 + \sigma_\eta^2 & \text{if } i = j, \\ \sigma_\eta^2 \alpha^{|T_i - T_j|} & \text{if } i \neq j, G_i = G_j, \\ 0 & \text{otherwise.} \end{cases}$$

MULTIPLE GROUPS AND MULTIPLE TIME PERIODS

- In the general case (multiple groups, multiple periods) the specification of the error components implies a specification of the full covariance matrix

$$\Omega = \mathbb{E}[\varepsilon\varepsilon'] \quad \text{where } \mathbf{Y} = \mathbf{X}\beta + \varepsilon,$$

- The variance of the OLS estimator is

$$\mathbb{V} \left(\hat{\beta}_{OLS} \right) = (\mathbf{X}'\mathbf{X})^{-1} (\mathbf{X}'\Omega\mathbf{X}) (\mathbf{X}'\mathbf{X})^{-1}.$$

- Alternatively one can do GLS, which improves the variance to

$$\mathbb{V} \left(\hat{\beta}_{GLS} \right) = \left(\mathbf{X}'\Omega^{-1}\mathbf{X} \right)^{-1}.$$

- Here focus on OLS estimator, because typically GLS does not improve precision much. Recall the clustering case where GLS had no benefits in the case with equal sized clusters and homoskedasticity, see Kloek (1981).

NONLINEAR EXTENSIONS: THE CHANGES IN CHANGES MODEL WITH CONTINUOUS OUTCOMES

- Return to the setting with two groups and two time periods. With the Meyer et al data the DID regression in levels gave us

$$\widehat{\text{duration}}_i = 6.27 + 4.91\widehat{\text{gr}}_i + 0.77 \times \widehat{\text{ti}}_i + 0.95 \times \widehat{\text{gr} \times \text{ti}}_i,$$

(0.52) (0.81) (0.76) (1.17)

- If we first transform the outcome to logarithms before doing the DID regression, we find

$$\log \widehat{\text{duration}}_i = 1.13 + 0.26 \times \widehat{\text{gr}}_i + 0.01 \times \widehat{\text{ti}}_i + 0.19 \times \widehat{\text{gr} \times \text{ti}}_i,$$

(0.03) (0.05) (0.04) (0.07)

- How do we compare / think about these differences?

NONLINEAR EXTENSIONS: THE CHANGES IN CHANGES MODEL WITH CONTINUOUS OUTCOMES

- Now these two models are **not** directly comparable.
- They make assumptions about independence of residuals in linear regressions based on different transformations of the dependent variables.

item As a result the two regressions **cannot** both be valid (unless the distributions of outcomes in the first period in both groups and in the second period control group are identical).

Theory typically does not tell us what the functional form should be.

NONLINEAR EXTENSIONS: THE CHANGES IN CHANGES MODEL WITH CONTINUOUS OUTCOMES

- Here we discuss an alternative approach, generalizing the standard DID model in several ways.
- First, we assume that in the absence of the intervention, the outcomes satisfy

$$Y_i(0) = h(U_i, T_i),$$

with $h(u, t)$ strictly increasing in u .

- The distribution of U_i is allowed to vary across groups, but not over time within groups, so that a key assumption is that

$$U_i \perp T_i \left| G_i.$$

NONLINEAR EXTENSIONS: THE CHANGES IN CHANGES MODEL WITH CONTINUOUS OUTCOMES

- The standard DID model embodies three additional assumptions, namely

$$U_i - \mathbb{E}[U_i|G_i] \perp G_i \quad (\text{additivity})$$

$$h(u, t) = \phi(u + \delta t), \quad (\text{single index model})$$

for a strictly increasing function $\phi(\cdot)$, and

$$\phi(\cdot) \text{ is the identity function.} \quad (\text{identity transformation})$$

- These three assumptions combined lead to

$$Y_{it}(0) = U_i + \delta t,$$

which gets us back to the standard DID model.

NONLINEAR EXTENSIONS: THE CHANGES IN CHANGES MODEL WITH CONTINUOUS OUTCOMES

- Identification requires that we can infer the distribution of $Y(0)|G_i = 1, T_i = 1$ from the distributions of $Y_i(0)|G_i = g, T_i = t$ for $(g, t) \neq (1, 1)$.
- The distribution of $Y(0)|G = 1, T = 1$ is **identified** under these assumptions.
- Let $F_{Y(0),gt}(y)$ denote the distribution function of $Y(0)$ given $G = g$ and $T = t$. Then

$$F_{Y(0),11}(y) = F_{Y(0),10} \left(F_{Y(0),00}^{-1} \left(F_{Y(0),01}(y) \right) \right).$$

- The distribution function of $Y(0)|G = T = 1$ can be estimated by plugging in empirical distribution functions for the other distribution functions in this expression.
- This can then be used in turn to calculate the mean of $Y(0)$ given $G = T = 1$ and the average effect of the new regime.

NONLINEAR EXTENSIONS: THE CHANGES IN CHANGES MODEL WITH DISCRETE DATA

- In settings with discrete outcomes the standard DID model is not very attractive.
- For example, with binary outcomes the assumption that in the absence of the treatment the outcomes would satisfy a linear additive model

$$\mathbb{E}[Y(0)|T, G] = \alpha_0 + \alpha_1 T + \alpha_2 G,$$

is very unattractive.

- It may violate the restriction that the expected value of a binary variable is between 0 and 1.

NONLINEAR EXTENSIONS: THE CHANGES IN CHANGES MODEL WITH DISCRETE DATA

- In this discrete case, the strict monotonicity assumption of the nonlinear model is also very restrictive.
- In that case a more plausible approach is to assume weak monotonicity:

$$h(u, t) \quad \text{is non-decreasing in } u.$$

- This model is **not** identified.
- We need an additional assumption.
- One such assumption that would be trivially satisfied in the strict monotonicity case is the following:

$$U \perp G \mid Y, T.$$

- Let us see how this works out in the binary outcome case.

NONLINEAR EXTENSIONS: THE CHANGES IN CHANGES MODEL WITH DISCRETE DATA

- Formally

$$\mathbb{E}[Y(0)|G = 1, T = 1] = \frac{\mathbb{E}[Y(0)|G = 0, T = 1]}{\mathbb{E}[Y(0)|G = T = 0]} \times \mathbb{E}[Y(0)|G = 1, T = 0]$$

$$\text{if } \frac{\mathbb{E}[Y(0)|G = 0, T = 1]}{\mathbb{E}[Y(0)|G = T = 0]} \leq 1$$

$$= 1 - \frac{1 - \mathbb{E}[Y|G = 0, T = 1]}{1 - \mathbb{E}[Y(0)|G = T = 0]} \times (1 - \mathbb{E}[Y(0)|G = 1, T = 0])$$

$$\text{if } \frac{\mathbb{E}[Y(0)|G = 0, T = 1]}{\mathbb{E}[Y(0)|G = T = 0]} > 1$$

REFERENCES

- ARKHANGELSKY, DMITRY, AND GUIDO IMBENS. CAUSAL MODELS FOR LONGITUDINAL AND PANEL DATA: A SURVEY. NO. W31942. NATIONAL BUREAU OF ECONOMIC RESEARCH, 2023.
- ATHEY, SUSAN, AND GUIDO W. IMBENS. "IDENTIFICATION AND INFERENCE IN NONLINEAR DIFFERENCE-IN-DIFFERENCES MODELS." *Econometrica* 74, NO. 2 (2006): 431-497.
- BERTRAND, MARIANNE, ESTHER DUFLO, AND SENDHIL MULLAINATHAN. "HOW MUCH SHOULD WE TRUST DIFFERENCES-IN-DIFFERENCES ESTIMATES?." *The Quarterly journal of economics* 119, NO. 1 (2004): 249-275.
- CARD, DAVID. "THE IMPACT OF THE MARIEL BOATLIFT ON THE MIAMI LABOR MARKET." *Industrial and Labor Relations Review* 43, NO. 2 (1990): 245-257.

REFERENCES (CTD)

- CARD, DAVID, AND ALAN B. KRUEGER. "MINIMUM WAGES AND EMPLOYMENT: A CASE STUDY OF THE FAST-FOOD INDUSTRY IN NEW JERSEY AND PENNSYLVANIA: REPLY." *American Economic Review* 90, NO. 5 (2000): 1397-1420.
- DEATON, ANGUS. "PANEL DATA FROM TIME SERIES OF CROSS-SECTIONS." *Journal of econometrics* 30, NO. 1-2 (1985): 109-126.
- HANSEN, CHRISTIAN B. "GENERALIZED LEAST SQUARES INFERENCE IN PANEL AND MULTILEVEL MODELS WITH SERIAL CORRELATION AND FIXED EFFECTS." *Journal of econometrics* 140, NO. 2 (2007): 670-694.
- KLOEK, TEUN. "OLS ESTIMATION IN A MODEL WHERE A MICROVARIABLE IS EXPLAINED BY AGGREGATES AND CONTEMPORANEOUS DISTURBANCES ARE EQUICORRELATED." *Econometrica* (1981): 205-207.
- MEYER, BRUCE D., W. KIP VISCUSI, AND DAVID L. DURBIN. "WORKERS' COMPENSATION AND INJURY DURATION: EVIDENCE FROM A NATURAL EXPERIMENT." *The American Economic Review* 85, NO. 3 (1995): 322-340.